

selectStrongestBbox

Select strongest bounding boxes from overlapping clusters

Syntax

```
[selectedBbox,selectedScore] = selectStrongestBbox(bbox,score)
[selectedBbox,selectedScore,index] = selectStrongestBbox(bbox,score)
[ __ ] = selectStrongestBbox( __ ,Name,Value)
```

Description

`[selectedBbox,selectedScore] = selectStrongestBbox(bbox,score)` returns selected bounding boxes that have a high confidence score. The function uses nonmaximal suppression to eliminate overlapping bounding boxes from the `bbox` input. example

`[selectedBbox,selectedScore,index] = selectStrongestBbox(bbox,score)` additionally returns the index vector associated with `selectedBbox`. This vector contains the indices of the selected boxes in the `bbox` input.

`[ __ ] = selectStrongestBbox( __ ,Name,Value)` uses additional options specified by one or more `Name,Value` pair arguments.

Examples

[collapse all](#)

✓ Run Nonmaximal Suppression on Bounding Boxes Using People Detector

Load the pretrained aggregate channel features (ACF) people detector.

Try This Example

Copy Command 

```
peopleDetector = peopleDetectorACF();
```

Detect people in an image. Disable the default nonmaximal suppression used by the detector.

```
I = imread('visionteam1.jpg');
[bbox,score] = detect(peopleDetector,I,'SelectStrongest',false);
```

Run nonmaximal suppression with custom threshold.

```
I = imread('visionteam1.jpg');
[selectedBbox,selectedScore] = selectStrongestBbox(bbox,score,'OverlapThreshold',0.3);
```

Display the results.

```
I1 = insertObjectAnnotation(I,'rectangle',bbox,score,'Color','r');
I2 = insertObjectAnnotation(I,'rectangle',selectedBbox,selectedScore,'Color','r');

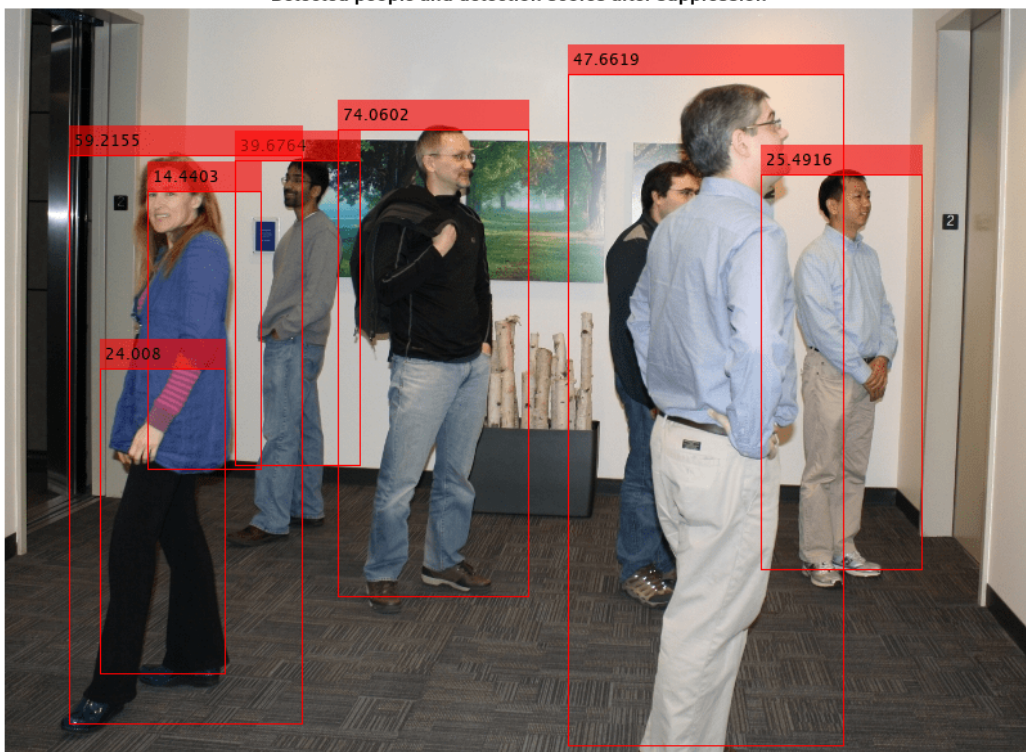
figure, imshow(I1);
title('Detected people and detection scores before suppression');
```

Detected people and detection scores before suppression



```
figure, imshow(I2);
title('Detected people and detection scores after suppression');
```

Detected people and detection scores after suppression

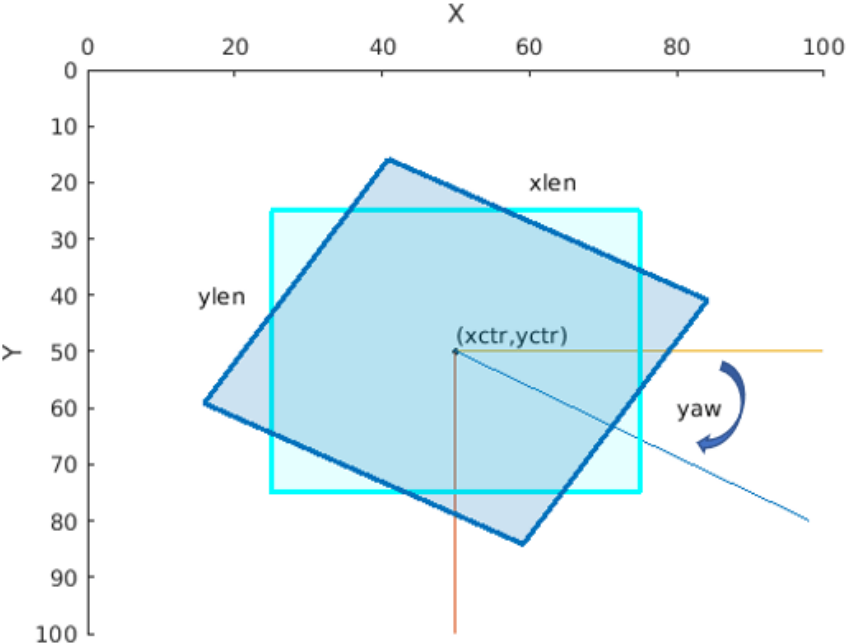


Input Arguments

[collapse all](#)

bbbox — Bounding boxes

Bounding boxes, specified as an *M*-by-4 or *M*-by-5 nonsparse numeric matrix. *M* is the number of bounding boxes. Each row of the matrix defines a bounding box as either an axis-aligned rectangle or a rotated rectangle. This table describes the format for each bounding box.

Bounding Box	Description
Axis-aligned rectangle	Defined in spatial coordinates as an <i>M</i>-by-4 numeric matrix with rows of the form $[x \ y \ w \ h]$, where: <i>M</i> is the number of axis-aligned rectangles. <i>x</i> and <i>y</i> specify the upper-left corner of the rectangle. <i>w</i> specifies the width of the rectangle, which is its length along the <i>x</i>-axis. <i>h</i> specifies the height of the rectangle, which is its length along the <i>y</i>-axis.
Rotated rectangle	Defined in spatial coordinates as an <i>M</i>-by-5 numeric matrix with rows of the form $[xctr \ yctr \ xlen \ ylen \ yaw]$, where: <i>M</i> is the number of rotated rectangles. <i>xctr</i> and <i>yctr</i> specify the center of the rectangle. <i>xlen</i> specifies the width of the rectangle, which is its length along the <i>x</i>-axis before rotation. <i>ylen</i> specifies the height of the rectangle, which is its length along the <i>y</i>-axis before rotation. <i>yaw</i> specifies the rotation angle in degrees. The rotation is clockwise-positive around the center of the bounding box. 

Data Types: single | double | int8 | int16 | int32 | uint8 | uint16 | uint32

▼ *M*-by-1 vector

Confidence score, specified as an *M*-by-1 vector. The *M*th score corresponds to the *M*th bounding box in the bbox input. The selectStrongestBbox function uses nonmaximal suppression to eliminate overlapping bounding boxes and associate the confidence score with the boxes. A higher score represents a higher confidence in keeping the bounding box. The score input must be real, finite, and nonsparse.

Data Types: single | double | int8 | int16 | int32 | uint8 | uint16 | uint32

Name-Value Arguments

Specify optional pairs of arguments as Name1=Value1, ..., NameN=ValueN, where Name is the argument name and Value is the corresponding value. Name-value arguments must appear after other arguments, but the order of the pairs does not matter.

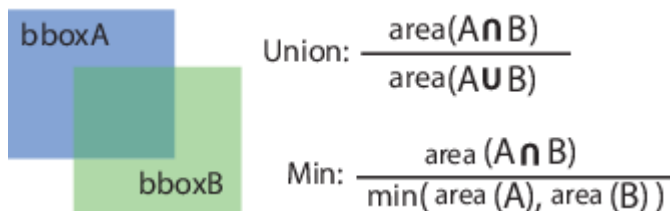
Before R2021a, use commas to separate each name and value, and enclose Name in quotes.

Example: 'RatioType','Union' sets the 'RatioType' property to 'Union'.

▼ **RatioType — Bounding box overlap ratio denominator**
'Union' (default) | 'Min'

Ratio type, specified as the character vector 'Union' or 'Min'.

- Set the ratio type to 'Union' to compute the ratio as the area of intersection between bboxA and bboxB, divided by the area of the union of the two.
- Set the ratio type to 'Min' to compute the ratio as the area of intersection between bboxA and bboxB, divided by the minimum area of the two bounding boxes.



Data Types: char

▼ **OverlapThreshold — Overlap ratio threshold**
0.5 (default) | scalar in the range [0 1]

Overlap ratio threshold, specified as the comma-separated pair consisting of 'OverlapThreshold' and a scalar in the range [0 1]. When the overlap ratio is above the threshold you set, the function removes bounding boxes around the reference box. Decrease this value to reduce the number of selected bounding boxes. However, if you decrease the overlap ratio too much, you might eliminate boxes that represent objects close to each other in the image.

Data Types: single | double

▼ **NumStrongest — Maximum number of strongest boxes**

inf (default) | positive scalar

Maximum number of strongest boxes, specified as the comma-separated pair consisting of 'NumStrongest' and inf or a positive scalar. Use this argument to reduce processing time when you have a priori knowledge about the maximum number of boxes. Set the value to inf to select all the strongest, non-overlapping, bounding boxes.

Output Arguments

[collapse all](#)



selectedBbox — Selected bounding boxes

M-by-4 matrix | *M*-by-5 matrix

Selected bounding boxes, returned as an *M*-by-4 or an *M*-by-5 matrix. The 4-element vectors represent axis-aligned rectangles and the 5-element vectors represent rotated rectangles.

The selectedBbox output returns the selected bounding boxes from the bbox input that have the highest confidence score. The function uses nonmaximal suppression to eliminate overlapping bounding boxes.

Data Types: single | double | int8 | int16 | int32 | uint8 | uint16 | uint32



selectedScore — Scores of selected bounding boxes

M-by-1 vector

Scores of selected bounding boxes, returned as an *M*-by-1 vector. The *M*th score in the selectedScore output corresponds to the *M*th bounding box in the selectedBbox output.

Data Types: single | double | int8 | int16 | int32 | uint8 | uint16 | uint32



index — Index of selected bounding boxes

M-by-1 vector

Index of selected bounding boxes, returned as an *M*-by-1 vector. The index vector contains the indices to the selected boxes in the bbox input.

Data Types: double

Extended Capabilities

› C/C++ Code Generation

Generate C and C++ code using MATLAB® Coder™.

› GPU Code Generation

Generate CUDA® code for NVIDIA® GPUs using GPU Coder™.

› GPU Arrays

Accelerate code by running on a graphics processing unit (GPU) using Parallel Computing Toolbox™.

Version History

Introduced in R2014b

See Also

`bboxOverlapRatio` | `selectStrongestBboxMulticlass`